Early-Warning Analytics for Bank Financial Health Using Multi-Year Financial Data

BankPulse AI — Financial Health Intelligence and Advisory Reporting

Team number	Team 1 (T01)
Project title	Bankpulse AI - A 360° view of banking health
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Deliverable code	D1 — Project Charter (8%)
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Course / Faculty guide	Analytics Capstone Project (ANCP) / Briti

1. Problem Statement

Context. Banks publish financial statements and ratios every period, but reviewing each ratio alone can hide unusual *combinations* of change across profitability, asset quality, capital, liquidity and efficiency. Analysts need a systematic, reproducible way to find such patterns, compare banks with peers and prioritize what to investigate (CRISP-DM; Wirth & Hipp, 2000).

> **Analytics question:** Using 5–10 years of Bloomberg ratios for 10–20 banks, can a financial analyst rank the top 10% most unusual bank-year observations with an Isolation Forest (Liu et al., 2008), compared with a robust statistical baseline on the same observations and features, so that at least 70% of flagged observations stay stable under reasonable parameter changes and 100% of flags and ratios are traceable to recorded inputs?

Decision-maker: a financial analyst or bank risk-analysis professional. **Decision supported:** which bank-year observations require further investigation because of unusual financial movements or combinations of indicators.

Output: a structured advisory report (KPI tables, trend charts, anomaly scores, explanations, recommendations). The system supports analyst review only; it does not declare a bank distressed, predict failure without outcome labels, or replace professional judgement.

2. Proposed Solution and Deliverables

BankPulse AI turns verified analytical outputs into a reviewable advisory report. Python calculates every number and draws every chart; GPT only interprets those verified results and never calculates or invents figures.

Table 1. *Role of each layer in the workflow.*

Layer	Role
Bloomberg	Supplies permitted bank financial statements and ratios
Python and pandas	Clean data, calculate indicators and peer differences, generate charts
Statistical baseline and Isolation Forest	Flag unusual bank-year observations on the same features
GPT	Explains which indicators drove each flag and drafts interpretations and recommended checks from verified outputs
Financial analyst	Reviews every flag and decides what to investigate

Table 2. *Components of the advisory report.*

Component	Content
Executive summary	One page: financial health, key changes, warning signals, areas to investigate
Performance charts	ROA, ROE, net interest margin; gross and net NPA; capital adequacy; loan, deposit and asset growth
Comparison tables	Banks against peers and sector medians: relative strengths, weaknesses, unusual movements
Early-warning signals	Anomaly flags with contributing indicators; a flag is not proof of distress
Advisory recommendations	Follow-up checks tied to observed data, marked for analyst review, not investment advice

Deliverables. Primary: an automatically generated PDF advisory report. Optional: an interactive dashboard (Power BI or Python). Supporting: cleaned data, calculated ratios, peer comparisons, model flags and source references for auditability, subject to Bloomberg licensing.

3. Stakeholder Map

Table 3. *Who decides, who is affected, and who supplies data.*

Stakeholder	Relationship	Role and interest
Financial / risk analyst	**Decides**	Primary user; decides which observations to investigate
Faculty guide (Briti)	**Decides** (project)	Reviews feasibility, rigour and milestone progress
Bank management and risk teams	**Affected**	Potential beneficiaries of clearer monitoring; may be subjects of flagged observations
Capstone team	**Builds**	Acquires and analyzes data, evaluates models, builds the reporting workflow
Bloomberg / institutional data provider	**Supplies data**	Provides permitted financial data and field definitions under licence

4. Success Metrics and Acceptance Criteria

Targets are proposed, not results, and will be confirmed after the data audit (D2); any infeasible target will be documented. Without independent ground-truth labels, anomaly detection cannot establish predictive accuracy or prove distress.

Table 4. *Target values and the baseline each is compared against.*

Metric	Target / acceptance criterion	Baseline
Data quality	At least 95% completeness of the final feature set (if feasible); missingness documented	Raw extracted dataset
Reproducibility	100% of reported ratios and flags traceable to recorded inputs, calculations and model settings	Manual traceability audit
Model comparison	Baseline and candidate scored on the same eligible observations and features	Robust statistical anomaly rules
Detection stability	Overlap of top-10% flags under reasonable parameter changes; target at least 70% if sample supports it	Primary candidate configuration
Interpretation	Every priority flag lists supporting indicators, period and limitations	Manual evidence audit
Reporting workflow	Complete report (charts, KPI tables, findings, limitations) generated from a versioned dataset	Manual report preparation

5. Scope and Assumptions

Table 5. *Scope boundaries and working assumptions.*

In scope

- Multiple banks over multiple financial years
- Bloomberg statements and ratios (permitted use)
- Cleaning, missing-value analysis, data dictionary, versioning
- EDA, trend analysis, peer benchmarking
- Statistical baseline and Isolation Forest candidate
- Evaluation, robustness checks, limitations
- AI-assisted advisory report (optional dashboard); deployment and monitoring plan

Out of scope

- Automated credit ratings or definitive failure predictions
- Personalized investment advice or buy/sell signals
- Autonomous financial or regulatory decisions
- Causal claims from observational data
- Redistributing Bloomberg data without authorization
- Production deployment beyond the team's access

Assumptions

- Institution permits academic use of required Bloomberg fields
- Enough comparable bank-year observations exist
- Team can use Python, pandas and ML libraries
- Ratios can be standardized, or differences documented
- Human review stays in the advisory process

6. Data Plan

Sources and access rights. Primary: Bloomberg Terminal statements and ratios (Bloomberg L.P., 2026), subject to institutional licensing and export permissions. Supplementary, only to validate definitions, investigate selected observations or fill documented gaps: public annual reports and official regulatory publications.

Table 6. *Candidate variables by dimension.*

Dimension	Candidate variables
Profitability	ROA, ROE, net interest margin
Asset quality	Gross and net NPA ratios, provisions relative to loans
Capital and liquidity	Capital adequacy and Tier 1 ratios; liquidity coverage ratio (where available and comparable)
Efficiency and growth	Cost-to-income ratio; loan, deposit and total asset growth
Derived features	Year-on-year changes; differences from peer medians

Expected size. About 10–20 banks over 5–10 years, i.e. roughly 50–200 bank-year observations. This is a feasibility estimate, not a confirmed size; it depends on field availability, consistency and completeness.

Privacy and consent. The data is institutional, not personal, so no individual consent is needed and DPDP Act, 2023 consent duties are not expected to apply; any change will be documented. The team will follow Bloomberg access and redistribution terms, maintain a source log and data dictionary, restrict access to permitted members, and keep credentials and unauthorized data out of the repository.

7. Feasibility and Top Five Risks

Table 7. *Top five risks with impact and mitigation.*

#	Risk	Potential impact	Mitigation
1	**Incomplete or inconsistent data**	Weak comparisons; unreliable features	Audit fields early, standardize definitions, document exclusions
2	**Small sample size**	Unstable anomaly detection	Start with transparent statistical rules; test whether the candidate model is justified
3	**No verified outcome labels**	Cannot validate actual distress prediction	Frame as anomaly detection; avoid unsupported distress claims
4	**Bloomberg access or licensing limits**	Data cannot be exported or shared as planned	Confirm institutional permissions; use authorized supplementary sources
5	**Misleading GPT interpretations**	Unsupported or fabricated explanations	Generate text only from verified outputs, trace to evidence, require human review